Flood Evacuation Route Recommendation using *Particle Swarm Optimization Algorithm*

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***Abstract*—Equipping the society with an evacuation route assistant is important as a preventive action in order to minimize anxiety as well as number of victims when disaster strike. Therefore, research on best route determination to the shelter is a noble task. Evacuation routes mean routes to pass in order to reach the gathering point or shelter safely. This study aims to implement the particle swarm optimization (*PSO*) algorithm in finding the optimal path to the designated shelter. PSO is a metaheuristic algorithm inspired by the behaviour of flocks of birds or fish to optimize the search for the best route. Taking into account the details of the PSO implementation, including initialization of the random particle position and velocity, selection of the best particle (*Pbest*) and Global Best (*Gbest*), and updating the speed and position of the particle using the specified formula. The objective function is recalculated based on the total distance of the particle route as well as the condition weight of the affected areas. The results showed that out of ten experiments with 100 iterations, six valid routes were found with the highest fitness value of 1,935, which indicates that PSO can provide optimal solutions in the context of finding flood evacuation routes**.

Keywords----*Route optimization, PSO, speed, location, assistant*

I. INRODUCTION

Natural disasters often occur in Indonesia. According to data from National Board for Disaster Management of Indonesia (Indonesian abbreviated as BNPB), between January to October 2022 there were 3,038 incidents, with 1,246 flood incidents as the most. Factors such as erratic weather changes, high rainfall, changes in land use, and poor drainage are the main causes of flooding. To reduce the number of victims, the competent authority is currently managing mitigation procedures and establishing evacuation gathering points close to settlements so that they can be easily accessed by the community [1].

Makassar City, as the capital city of South Sulawesi province, has a dense population reaching 1,423,877 people in 2020, with a population density of 8,101 people/Km2. Based on the report of the Indonesian Disaster Risk Index (Indonesian abbreviated as IRBI) [2], Makassar City is generally categorized as a city with a moderate disaster risk index. IRBI also reported that Makassar City has a different level of risk index for each type of threat, such as flooding 12.00 (moderate), tsunami 24.00 (high), earthquake 10.40 (medium), forest and land fires 24.00 (high), high waves of absorption 24.00 (high), extreme weather 13.60 (high), landslides 12.00 (moderate), and drought 24.00 (high).

The regional government, in this case the Makassar City Regional Disaster Management Agency (Indonesian abbreviated as BPBD), has not a system yet to help people to determine the best route to the evacuation assembly point that the community can use when disaster strikes [3]. When a disaster occurs, the panicked state of residents who do not have clear information about evacuation points and evacuation routes causes the evacuation process to be slower or even difficult. Evacuation situations that are slow and irregular, accompanied by uncontrolled panic among disaster victims, can increase the number of deaths.

In this study, we implement the Particle Swarm Optimization (*PSO*) algorithm as an optimization method to solve problems in determining the best route to the shelter when facing natural disasters, especially floods in Makassar City. PSO has been consider powerful in solving complex problem. As PSO has the ability to perform a global search in the solution space, which means finding the best solution among many possible solutions. Fig. 1 presents how the flow of the PSO Algorithm, which started with parameter adjustment following by initiating particles or data, continuing by couple of iterations with series calculation to find satisfied criteria, and finally ends with testing data to validate the result. In the context of determining the best route to the shelter, PSO can determine an efficient and safe route for the community when facing a flood disaster [4]. In addition, PSO has a good flexibility, so it can adapt to changing conditions and requirements. In the context of natural disasters, which often involve sudden changes in road conditions and accessibility, the ability of PSOs to adapt can help to determine the best route d dynamically [5].

Based on these explanation, the research question is how powerful the Particle Swarm Optimization Algorithm for flood evacuation route recommendation. This research question will be investigated by conducting a couple of experiments to find the solution.

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II. RESEARCH METHODS

The PSO method is a simple optimization method for changing some parameters, by selecting attributes or features and increasing attribute weights on all variables or attributes used [6]. This algorithm is inspired by the behavior of birds flying or fish swimming in schools. Despite being in very close proximity, hundreds of birds or fish can move quickly and efficiently without colliding. The main advantage of PSO is that it is simple, easy to implement, and has a high convergence speed [7]. PSO also can be used for raw data or opinions in sentences to convert them into numeric form by combining the TF-IDF method to convert text into numeric data.

Searching for solutions in the PSO algorithm by a certain population based on a number of particles. The population is assessed randomly and has the smallest and largest value limits. Particles track solutions by going through the search space by adapting to their best location (*local best*) and adapting to the best particle locations in all groups (*global best*) while going through the search space [8].

This research refers to the condition of Manggala district, an area in Makassar city South Sulawesi, Indonesia as a case study, which is experiencing quite high intensity of flood disasters. The data used in this research is the area status maps, which are divided into very safe, safe, vulnerable, very vulnerable and alert areas as the basis for determining the route to the evacuation location. In implementing the method, each road intersection is identified as a node, which is then used to determine the path. These road intersections are represented in the form of interconnected coordinate points. Data on affected areas as well as intersection points are the key parameters in this research.

A. Prelimenary Data and Scenario

As an effort to improve recommended evacuation routes, the Particle Swarm Optimization (PSO) algorithm was used in designing the initial scenario. In this situation, there are two shelters or evacuation sites located around the flood-affected areas [1]. In the scenario shown in Fig. 2, there are roads which are symbolized by J1 to J12, while the crossing points or junction between road segments are symbolized by D1 to D6 and added by points S1 and S2 as shelter points. Each road segment has a distance recorded in Table I.

In addition, the condition of the road affected by the flood also affects the determination of the evacuation route [10]. To accommodate the impact of flooding, weighting is carried out on the road conditions listed in Table II. Weighted scores, which range from 0.1 to 1, are assigned for each road condition. This score reflects the severity of the impact of flooding on the roads.

Furthermore, the distance data from Table I and weighted scores from Table II are then applied to the Particle Swarm Optimization (PSO) algorithm. This algorithm is used to evaluate the success rate in determining evacuation routes when a flood occurs [11]. The PSO algorithm utilizes the connection information between points listed in Table III to find the optimal solution that takes into account the distance, road conditions and intersection points.

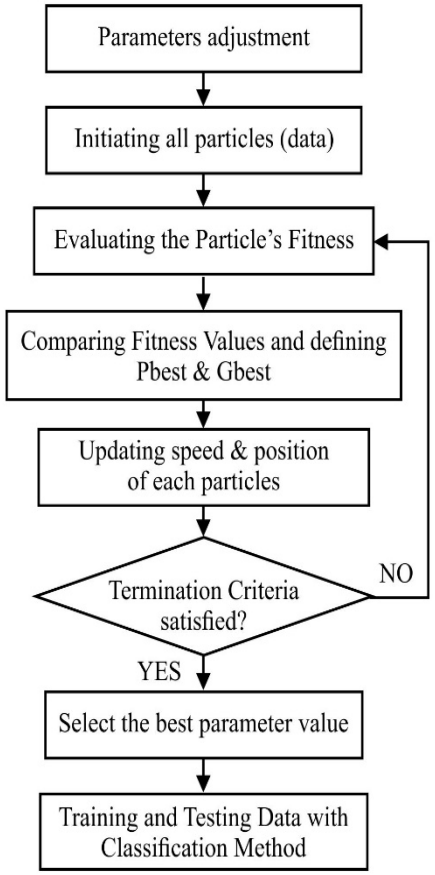


Fig. 1. PSO Implementation Process [9]

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Thus, through a combination of distance data, road condition weighting, and the use of the PSO algorithm, an efficient and safe evacuation route can be found in the case of flood disaster. Fig. 2 presented the research scenario with routes that have been grouped by distance, road conditions and crossing points. It can be seen in Fig. 2 that although the shortest path is D1-D5-S1, this path will not be eligible for recommendation route, since this path crossing the alert areas indicated by the red line. Concerning the status of the road presented in the scenario, possible route to be recommended are paths heading to shelter 1 which are in safe and very save status. While paths heading to shelter 2 are mostly in alert and very vulnerable status. These situations are going to be solved by implementing the PSO algorithm as discussed in the rest of this paper.

TABLE 1. DISTANCE OF EACH ROAD SELECTION

Road	Distance (M)
J1	20
J2	10
J3	20
J4	20
J5	20
J6	20
J7	20
J8	10
J9	10
J10	10
J11	10
J12	40

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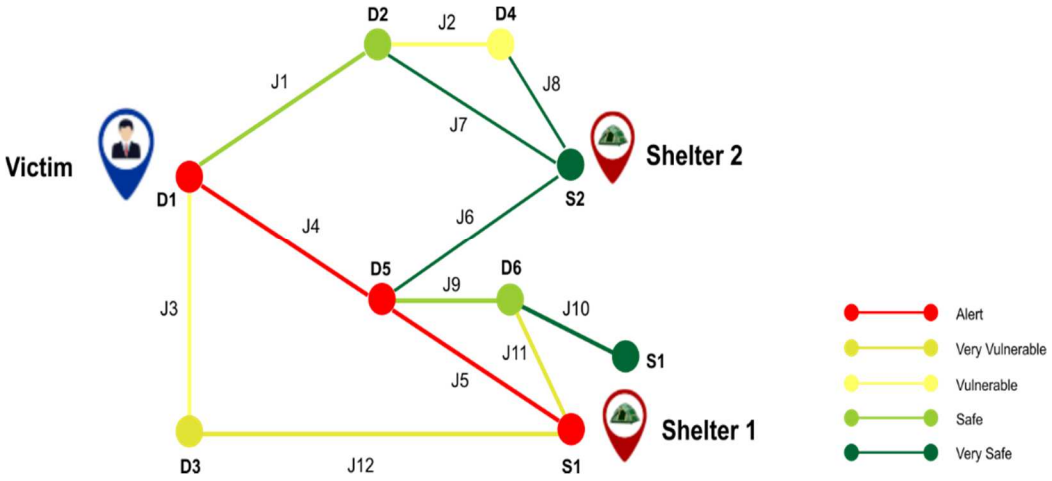


Fig. 2. Route and Road Condition

TABLE II. ROAD WEIGHTING

Road Condition	Weight
Alert	0,1
Very Vulnerable	0,5
Vulnerable	0,75
Safe	0,9
Very Safe	1

TABLE III. CROSSING POINTS

Crossing Points/ Junction	Connection
D1	D2,D3,D5
D2	D1,D4,S2
D3	D1,S1
D4	D2,S2
D5	D1,D6,S2,S1
D6	D5,S1
S1	D3,D5,D6
S2	D2,D4,D5

B. *Implementation of The PSO Algorithm*

There are seven steps in implementing the Particle Swarm Optimization (PSO) algorithm in this study, as follows:

1) *Initialize the particle's position and velocity*. At this stage, the particle position and velocity are randomly initialized [12]. In this study, there are 5 particles, and the symbol D1 has a value of 0.1 which is used as a starting point or user point, can be seen in Table IV. Then the objective function (*fitness*) is calculated based on the initial position.

2) Choose the best particle (*Pbest*) and the Best Global (*Gbest*). Each particle has a Pbest value which represents its best position. In addition, Gbest is the best position found globally for all particles [13].

3) Particle speed update. At this stage, the weighting factor (p) is set to 0.9, the value of r^1 1 is 0.4 and the value of r^2 is 0.6. Then the particle velocity is updated using the following formula [14].

$$v = p * v + r^1 * (\text{Pbest} - x) + r^2 * (\text{Gbest} - x)$$

Where:

- v is the particle speed
- p is the weighting factor
- r^1 and r^2 is a random value that affects the speed change
- Pbest is best position of the particle
- Gbest is the best position globally
- x is the current position of the particle

4) *Particle position update*. The particle position is updated using the following formula [15].

$$x = x + v$$

Where:

- x is the particle position
- v is the updated the particle velocity

TABLE IV. INITIALISATION OF PARTICLE'S POSITION AND VELOCITY

	D1	D2	D3	D4	D5	D6	S1	S2
Particle 1	0,1	0,641533	0,654688	0,243441	0,604383	0,794709	0,731051	0,07269
Particle 2	0,1	0,08283	0,30199	0,156822	0,72979	0,614547	0,174382	0,689668
Particle 3	0,1	0,466301	0,653559	0,039202	0,928026	0,980267	0,928466	0,262354
Particle 4	0,1	0,63427	0,230674	0,137408	0,736136	0,769387	0,453508	0,183571
Particle 5	0,1	0,606761	0,741608	0,044689	0,480543	0,042433	0,589268	0,929315

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5) Calculate new goal function (*fitness*).

After the particle positions are updated, the objective function (fitness) is recalculated for each particle using the following formula [16].

$$f = \left(\frac{1}{x}\right) * n$$

Where:

- f is the value of the objective function
- x is the total route distance of the particle
- n is the weight of the road condition of the affected area that is relevant to the route path

6) Update *Pbes*t and *Gbest* if needed

Pbest is updated for each particle if the new objective function value is better than before. In addition, Gbest is updated if there are particles with objective function values that are better than the current *Gbset* [17].

7) Repeat Steps 3 to 6 until it reaches the maximum iteration. Steps 3 to 6 are repeated for a maximum of 100 iterations to get the optimal route.

C. Evaluation

Evaluation were conducted in order to investigate the performance of the Particle Swarm Optimization (PSO) algorithm in solving the specified problem. Testing were carried out on the resulting evacuation routes [1] by reviewing valid routes resulting from 10 trials based on given scenario in Fig. 2.

III. RESULT AND DISCUSSION

Out of ten (10) experiments were carried out based on the results of the implementation of the Particle Swarm Optimization (PSO) algorithm. Six (6) trials were found that produced valid routes with the high fitness value of 1,935 while other four (4) routers consider invalid as can bee seen in Table V.

The experiment results in Table V were obtained using 5 particles in the PSO algorithm. The parameters used are the number of particles of 5, the weighting factor of 0.9, the value of r^1 is 0.4, the value of r^2 is 0.6 and the maximum number of iterations is 100. In this experiment, two valid route options were found, namely D1-D2-S2-D4-S2-D5-S1-D6-S1 and D1-D3-S1-D6-S1. During the experiment, it was revealed that the route D1-D2-S2-D4-S2-D5-S1-D6-S1 had a higher level of optimality with a fitness of 1.935. This route was successfully found in 6 out of 10 experiments, by obtaining a valid route D1-D2-S2. Although other route variations were existing, this route consistently shows high potential to be a more optimal route without any parameter changes in terms of the weighting conditions of the particular affected areas.

Furthermore, another important issue is the computation time. Through several experiments, it is observed that it takes around 89 second in average for the algorithm to present the recommendation result. This is because the system reads all existing intersections as illustrated in Fig. 3. Therefore, the more intersections, the slower the recommendation output will be. Therefore, further investigation is needed to deal with this issue.

TABLE V. EVALUATION RESULT

Esperiment	Optimal route	Rute Valid	Validity	Fitness Value
1	**D1-D2-S2**-D4-S2-D5-S1-D6-S1	**D1-D2-S2**	Yes	1.935
2	**D1-D3-S1**-D6-S1	**D1-D3-S1**	No	1.212
3	**D1-D3-S1**-D6-S1	**D1-D3-S1**	No	1.212
4	**D1-D2-S2**-D4-S2-D5-S1-D6-S1	**D1-D2-S2**	Yes	1.935
5	**D1-D2-S2**-D4-S2-D5-S1-D6-S1	**D1-D2-S2**	Yes	1.935
6	**D1-D3-S1**-D6-S1	**D1-D3-S1**	No	1.212
7	**D1-D2-S2**-D4-S2-D5-S1-D6-S1	**D1-D2-S2**	Yes	1.935
8	**D1-D2-S2**-D4-S2-D5-S1-D6-S1	**D1-D2-S2**	Yes	1.935
9	**D1-D2-S2**-D4-S2-D5-S1-D6-S1	**D1-D2-S2**	Yes	1.935
10	**D1-D3-S1**-D6-S1	**D1-D3-S1**	No	1.212

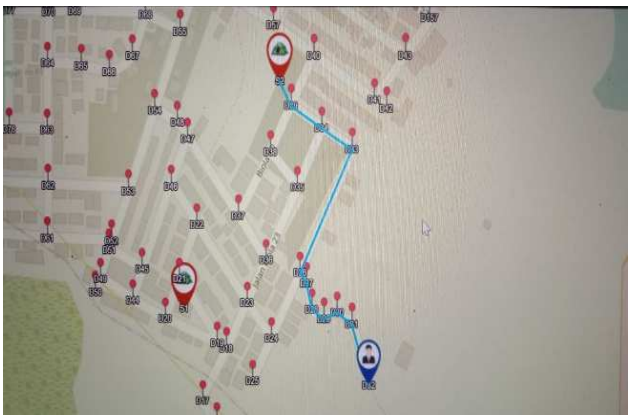


Fig. 3. Route Recommendation Visualization

IV. CONCLUSION

It is confirmed that the valid route for the given scenario is D1-D2-S2 with a fitness value of 1.935 from the optimal route D1-D2-S2-D4-S2-D5-S1-D6-D1 which found in 6 out of 10 trials. Based on the results of these experiments, the D1-D2-S2 route has a higher possibility of being the most optimal route (to be recommended) in terms of the weighting road conditions of the particular affected area.

Regarding the processing time issue, our future work will be focused on reducing the computation time by applying certain mechanism to reduces the number of intersections that must be considered by the algorithm during the computation process, such as filtering the involved intersection by zone clustering.

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